Plant Leaf Disease Classification using Deep learning based Hybrid Approach

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Abstract- Plant diseases lower crop quality and quantity, making them crucial. Thus, early disease identification and diagnosis are essential. To automate Maize leaf disease classification, we propose a deep learning-based approach. Convolutional Neural networks identify picture databases and perform best in image classification. We customized the Logistics Regression and applied the Convolutional Neural Network(hybrid) method to extract deep image characteristics. This research uses the maize leaves dataset from the Plant Village and PlantDoc datasets. The proposed method yields 93.2% accuracy in experiments.

Keywords: Corn or Maize leaf disease, hybrid model, covolutional neural network, Logistic Regression.

I. INTRODUCTION

Analysis of healthy and diseased plants is crucial in developing thriving agriculture. An essential step for defending uninfected plants against infected plants is the identification of the affected plants [1]. Most illness symptoms can be seen on plant leaves, making them the primary source for spotting leaf infections [2]. Recognizing various indications of various illnesses, leaf disease detection is the most advised method for detecting plant infection. Maize, also known as corn, can be affected by multiple leaf diseases, significantly impacting crop yield and quality. Grey Leaf Spot, Blight, and Common Rust are the most common maize leaf diseases. Prevention and control measures for maize leaf diseases include using disease-resistant varieties, rotating crops, practicing sound sanitation, and applying fungicides and other chemical treatments when necessary [3,4].

Automating disease identification reduces farm monitoring labor.[5] This study shows the best crop disease detection methods using deep-learning algorithms [5,6]. This paper compares Logistic regression (LR), Convolutional Neural Network (CNN), and ensemble deep learning NNLR for maize leaf disease detection algorithm regarding classification accuracy. A literature review, dataset, approach, performance calculation, results, discussion, and conclusion are as follows.

II. LITERATURE SURVEY

Identifying maize leaves is crucial in agriculture because it permits researchers to assess plant vitality and growth. Using variables such as the texture [7], type [8], and color [9] of plant leaf photos, most machine-learning studies have focused on the categorization of plant diseases. It stresses the significance of maize leaf detection in agricultural research and the need for more precise algorithms to enhance crop monitoring and plant health evaluation. Most previously used machine learning algorithms were designed for use in controlled environments. Therefore, they lack the resilience required for use in real-world agricultural settings. There has been a recent uptick in the use of deep learning (DL) techniques in agriculture, particularly those based on convolutional neural networks (CNNs), for a variety of detection and classification tasks, including weed detection [10], crop pest classification [11], and plant disease diagnosis [12]. Deep learning (DL) is a subfield of machine learning that has found widespread use in areas such as image classification, object identification, and NLP [12]. Deep learning (DL) is a subfield of ML. The issues of low performance [13], a lack of actual images [14], and segmented operation [15] that plagued conventional machine learning techniques have been addressed, at least in part, by this approach. One significant benefit of DL models is that they do not require segmented operations to extract features yet still achieve good results.

III. USED DATASET AND APPROACH

It has two subsections.

A. Dataset

This research uses Kaggle data set [16].

The datasets on maize are divided into four categories.

a) Corn blight (1146 images) identified a
b) Common rust (1306 images) identified as b
c) Grey speck (574 images) designated c
d) Healthy(1162 images) referred to as d

66% of the data in the collection is comprised of training data, while 34% is made up of assessment data. Figure 1 shows samples of each of these four types of images.

Fig. 1. Four types of leaves

B. Approach

The ensemble technique improves standard classifiers. Ensemble deep learning combines Neural Network and Logistic Regression classifiers and evaluates parameters.

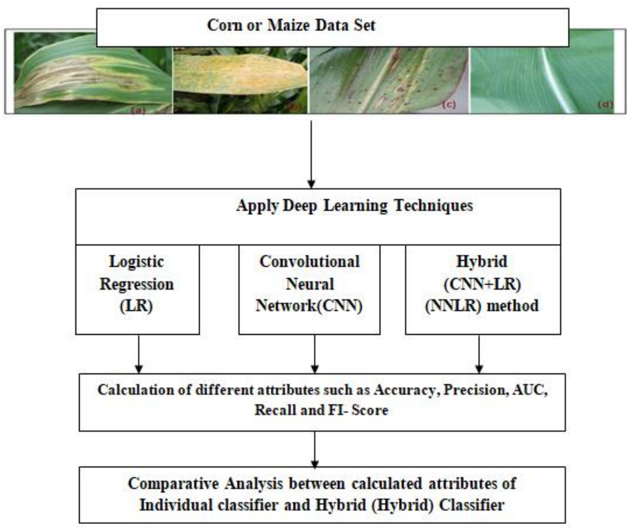

Fig. 2. Procedural Approach

Figure 2 illustrates leaf health. Identification strategy:

1. Dataset Acquisition
2. Deep learning classifiers like CNN, LR, and the Hybrid technique (NNLR) distinguish diseased and healthy leaf pictures.
3. Performance Measurements.
4. Result Comparisons.

Deep learning uses convolutional neural networks and logistic regression. NNLR is the ensemble approach presented. Hybrid (NNLR) technique calculates each parameter.

IV. Performance Calculation

Following Matrices are used for performance calculations. True positives (Te+, accurately anticipated positive values), false positives (F+, inaccurately forecasted positive values), true negatives (Te-, accurately predicted negative values), and false negatives (F-, inaccurately predicted negative values) make up a matrix.

A. Classification Accuracy(C)

$$(\mathrm{C}) = \frac{(Te+)+(Te-)}{((Te+)+(Te-)+(F+)+(F-)))} \qquad (1)$$

B. Precision (P)

$$(\mathrm{P}) = \frac{(\mathrm{Te+})}{(\mathrm{Te+})+(\mathrm{F-})} \qquad (2)$$

C. Recall(R)

$$(\mathrm{R}) = \frac{(\mathrm{Te+})}{(\mathrm{Te+})+(F-)} \qquad (3)$$

D. F 1 – Score

$$\frac{2*(\mathrm{R})*(\mathrm{P})}{(\mathrm{R})+(\mathrm{P})} \qquad (4)$$

V. Discussions and Results

Figures 3-6 illustrate the five-fold performance plot for corn blight, common rust, grey spot, and healthy pictures. Green indicates LR, orange CNN, and purple is the calibration plot's ensemble (NNLR) approach. Figures 7-10 illustrate the performance plot for tenfold corn blight, common rust, grey spot, and healthy pictures. Figures 11-14 demonstrate the twenty-fold performance plot for corn blight, common rust, grey spot, and healthy pictures.

TABLE I. Comparison at 20 K folds

Model	AUC	CA	F1	Precision	Recall
Hybrid model	0.9906	0.9322	0.9312	0.9311	0.9315
CNN	0.9885	0.9312	0.9308	0.931	0.9312
LR	0.9886	0.9241	0.9243	0.9243	0.9247

TABLE II. 20-fold hybrid model confusion matrix

		Predicted				
Actual		a	b	c	d	∑
	a	1040	18	82	6	1146
	b	26	1267	11	2	1306
	c	127	10	435	2	574
	d	3	1	0	1158	1162
	∑	1196	1296	528	1168	4188

TABLE III. CNN model's 20-fold confusion matrix

		Predicted				
Actual		a	b	c	d	∑
	a	1041	18	80	7	1146
	b	29	1261	15	1	1306
	c	123	9	440	2	574
	d	4	0	0	1158	1162
	∑	1197	1288	535	1168	4188

TABLE IV. The LR model's confusion matrix at 20K folds

		Predicted				
Actual		a	b	c	d	∑
	a	1024	18	99	5	1146

	b	28	1264	12	2	1306
	c	132	11	428	3	574
	d	4	1	0	1157	1162
	∑	1188	1294	539	1167	4188

TABLE V. COMPARISON AT 10K FOLDS

Model	**AUC**	**CA**	**F1**	**Precision**	**Recall**
Hybrid model	0.9901	0.9305	0.9302	0.9304	0.9305
CNN	0.9882	0.92908	0.92854	0.92859	0.9298
LR	0.9882	0.9221	0.9218	0.9217	0.9221

TABLE VI. 10-FOLD HYBRID MODEL CONFUSION MATRIX

	Predicted					
Actual		a	b	c	d	∑
	a	1035	17	87	7	1146
	b	27	1268	10	1	1306
	c	129	11	431	3	574
	d	4	1	0	1157	1162
	∑	1195	1297	528	1168	4188

TABLE VII. CNN MODEL'S 10-FOLD CONFUSION MATRIX

	Predicted					
Actual		a	b	c	d	∑
	a	1036	16	86	8	1146
	b	29	1262	14	1	1306
	c	125	6	440	3	574
	d	2	1	0	1159	1162
	∑	1192	1285	540	1171	4188

TABLE VIII. THE LR MODEL'S CONFUSION MATRIX AT 10K FOLDS

	Predicted					
Actual		a	b	c	d	∑
	a	1015	19	106	6	1146
	b	31	1264	10	1	1306
	c	131	13	427	3	574
	d	4	1	1	1156	1162
	∑	1181	1297	544	1166	4188

TABLE IX. COMPARISON AT 5K FOLDS

Model	**AUC**	**CA**	**F1**	**Precision**	**Recall**
Hybrid model	0.9895	0.9281	0.9273	0.9278	0.9281
CNN	0.9873	0.9274	0.9266	0.927	0.9274
LR	0.9873	0.9216	0.9212	0.9212	0.9216

TABLE X. 5-FOLD HYBRID MODEL CONFUSION MATRIX

	Predicted					
Actual		a	b	c	d	∑
	a	1046	16	79	5	1146
	b	28	1265	12	1	1306
	c	143	12	418	1	574
	d	4	0	0	1158	1162
	∑	1221	1293	509	1165	4188

TABLE XI. . CNN MODEL'S 5-FOLD CONFUSION MATRIX

	Predicted					
Actual		a	b	c	d	∑
	a	1043	20	74	9	1146
	b	29	1262	15	0	1306
	c	140	11	421	2	574
	D	4	0	0	1158	1162
	∑	1216	1293	510	1169	4188

TABLE XII. THE LR MODEL'S CONFUSION MATRIX AT 5K FOLDS

	Predicted					
Actual		a	b	c	d	∑
	a	1022	15	104	525	1146
	b	29	1262	15	2	1306
	c	139	14	419	2	574
	d	4	1	0	1157	1162
	∑	520	458	228	1206	4188

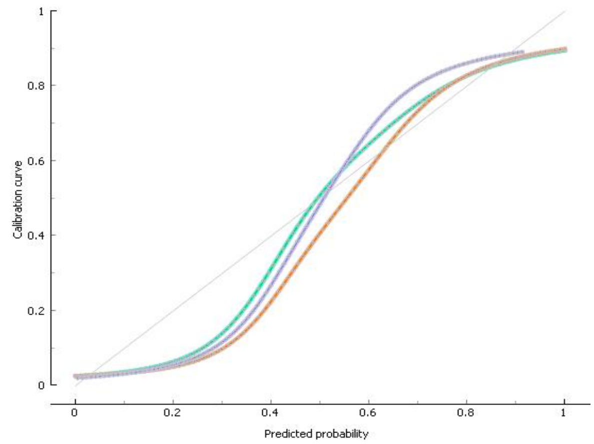

Fig. 3. . Calibration plot at 5 K folds blight

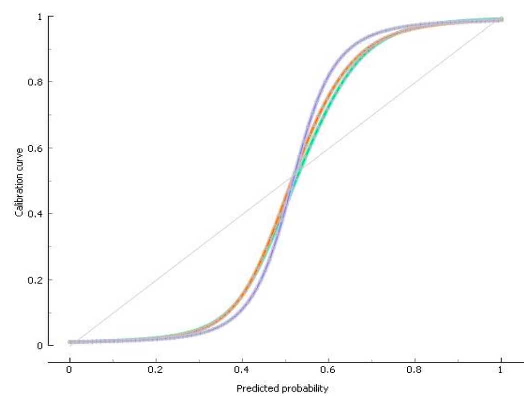

Fig. 4. Calibration plot at 5 K folds common rust

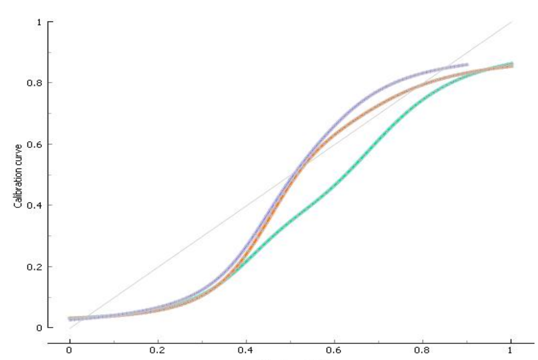

Fig. 5. Calibration plot at 5 K folds grey spot

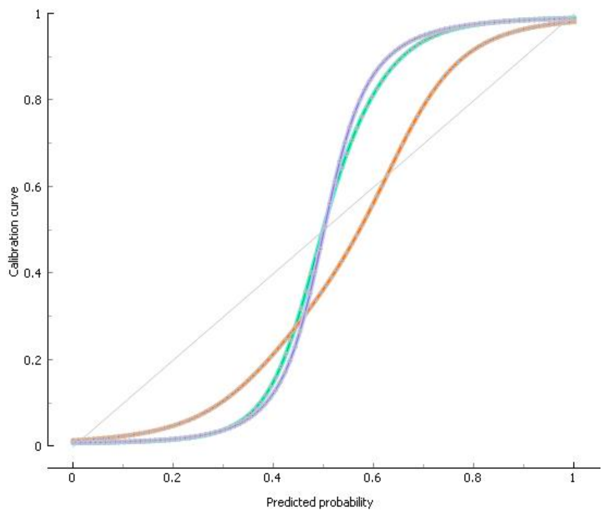

Fig. 6. Calibration plot at 5 K folds healthy

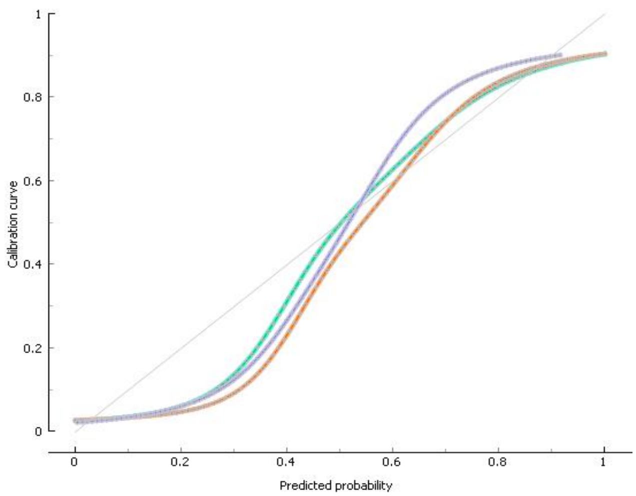

Fig. 7. Calibration plot at 10 K folds blight

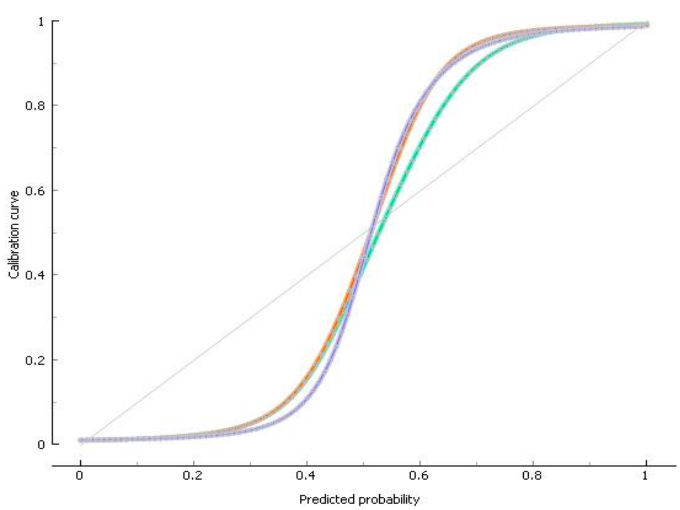

Fig. 8. Calibration plot at 10 K folds commonrust

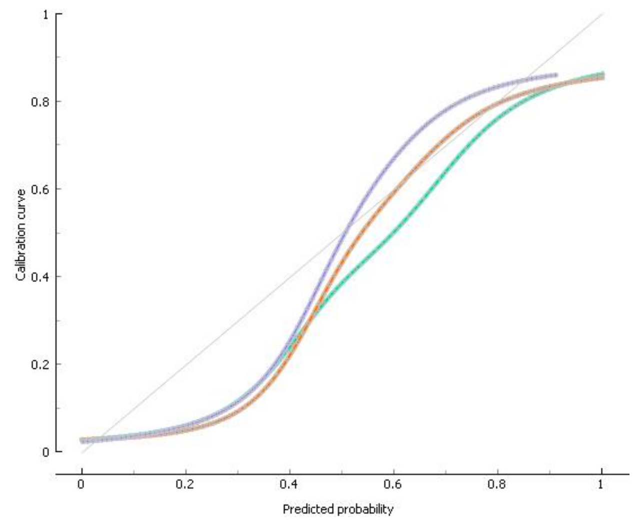

Fig. 9. Calibration plot at 10 K folds grey spot

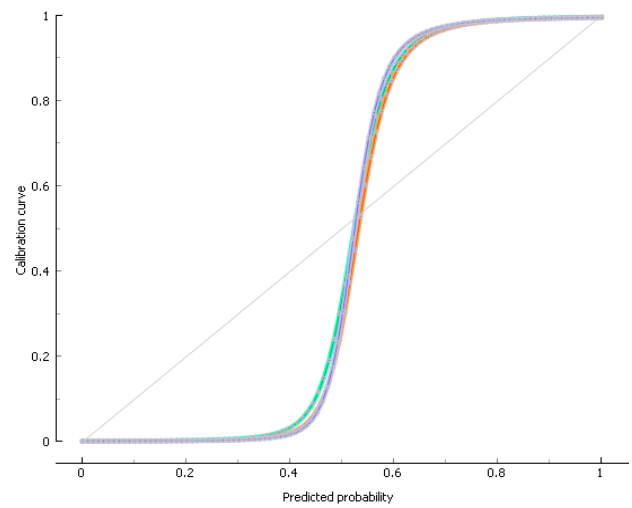

Fig. 10. Calibration plot at 10 K folds healthy

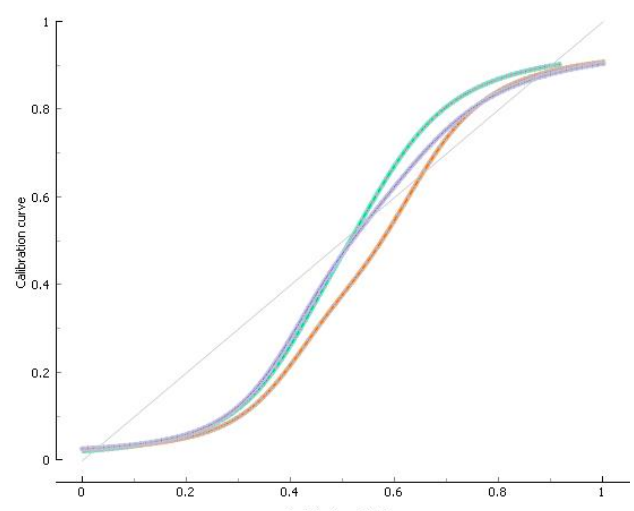

Fig. 11. Calibration plot at 20 K folds blight

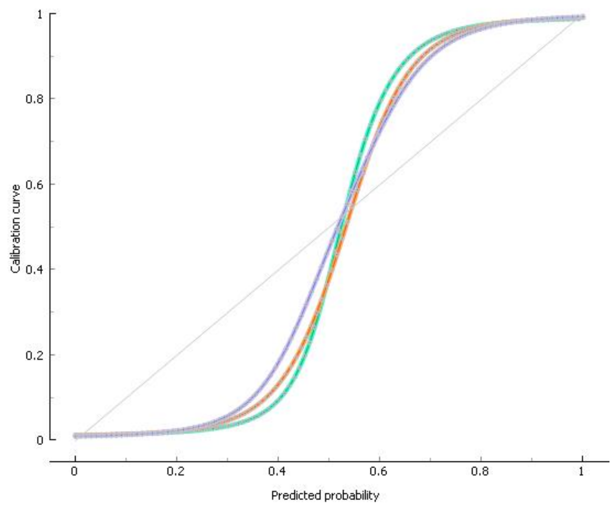

Fig. 12. Calibration plot Common rust at 20 K folds

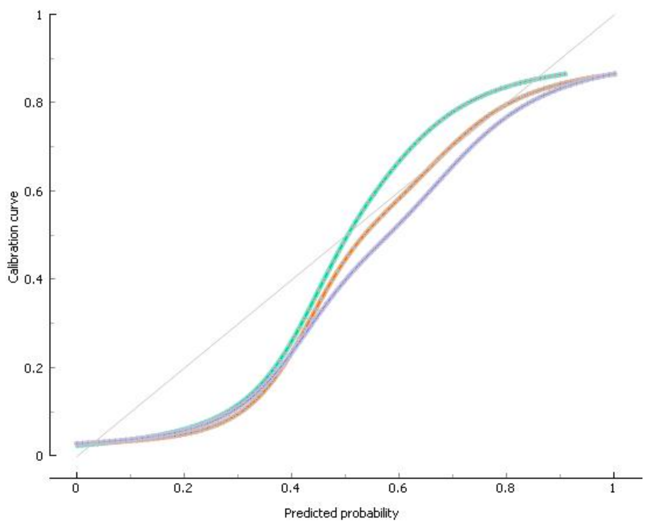

Fig. 13. Calibration plot at 20 K folds grey spot

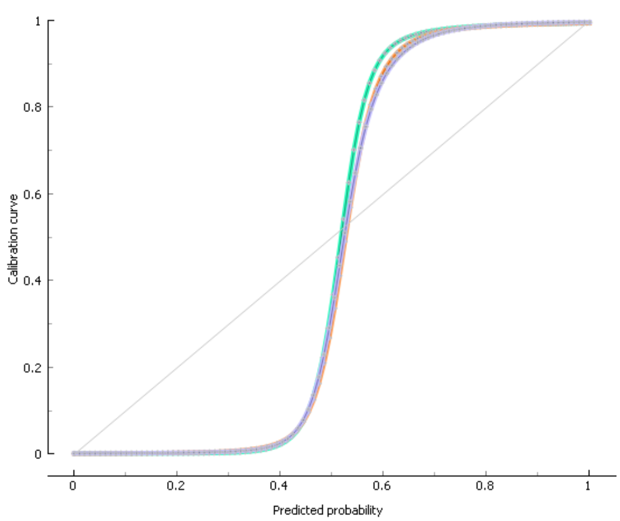

Fig. 14. Calibration plot at 20 K folds healthy

VI. CONCLUSION

In this study, we extracted and classified maize leaf disease features using CNN, LR, and NNLR models. Kaggle's "plant village dataset" provided maize leaf images.[5] Accuracy, recall, F1-score, and precision are calculated for five, ten, and 20 folds; Tables 1, 5, and 9 show that the ensemble approach (NNLR) improved all parameters. For 20-fold, 10-fold, and five-fold cross-validation, the hybrid (NNLR) technique achieves 93.2%, 93.05%, and 92.8% accuracy.

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